

Programming for Data Analysis Assignment Report Using Python and R

Fiha Akter
Department of AI & Data Science
Green University of Bangladesh

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1 Introduction

Data analysis is the process of collecting, organizing, exploring, and interpreting data to extract meaningful insights. Python and R are two of the most widely used programming languages for data analysis because they provide powerful tools for handling datasets, performing statistical analysis, and creating visualizations.

In this assignment, both Python and R were used to perform three basic data analysis tasks:

- Loading and exploring a CSV dataset.
- Creating new columns using existing data.
- Sorting the dataset based on a selected column.

These assignments provide a practical understanding of basic data manipulation techniques that are commonly used in real-world data analysis projects.

2 Dataset

The Iris dataset was used in this assignment.

Dataset Name: Iris Dataset

Dataset Source:

<https://raw.githubusercontent.com/mwaskom/seaborn-data/master/iris.csv>

The dataset contains five columns:

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width
- Species

The dataset consists of 150 observations and 5 variables

3 Assignment 1: Load and Explore Data (Python)

3.1 Objective

The objective of this assignment is to learn how to load a CSV dataset from the Internet and perform basic data exploration using Python and the Pandas library.

3.2 Python Code

```
1 import pandas as pd
2
3 # Load CSV dataset from the internet
4 url = "https://raw.githubusercontent.com/mwaskom/seaborn-data/
   master/iris.csv"
5
6 df = pd.read_csv(url)
7
8 # Display first 5 rows
9 print("First 5 Rows:")
10 print(df.head())
11
12 # Display column names
13 print("\nColumn Names:")
14 print(df.columns.tolist())
15
16 # Display number of rows and columns
17 print("\nDataset Shape:")
18 print("Rows:", df.shape[0])
19 print("Columns:", df.shape[1])
20
21 # Display summary statistics
22 print("\nSummary Statistics:")
23 print(df.describe(include='all'))
```

3.3 Output

First 5 Rows:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

Column Names:

```
['sepal_length', 'sepal_width',
 'petal_length', 'petal_width',
```

```
'species']
```

Dataset Shape:

Rows: 150

Columns: 5

Summary Statistics:

```
count 150
mean 5.843333
std 0.828066
min 4.300000
25% 5.100000
50% 5.800000
75% 6.400000
max 7.900000
```

3.4 Explanation

In this assignment, the Iris dataset was loaded directly from an online source using the `read_csv()` function of the Pandas library. The `head()` function displayed the first five rows of the dataset, while the column names were obtained using the `columns` attribute. The total number of rows and columns was determined using the `shape` attribute. Finally, the `describe()` function generated summary statistics for the numerical columns, including the count, mean, standard deviation, minimum, maximum, and quartile values. These steps provide an overview of the dataset before performing further analysis.

4 Assignment 1: Load and Explore Data (R)

4.1 Objective

The objective of this assignment is to learn how to load a CSV dataset from the Internet and perform basic data exploration using the R programming language.

4.2 R Code

```
1 # Load CSV dataset from the internet
2
3 url <- "https://raw.githubusercontent.com/mwaskom/seaborn-data/
4     master/iris.csv"
5
6 df <- read.csv(url)
7
8 # Display first 5 rows
9 cat("First 5 Rows:\n")
10 head(df, 5)
11
12 # Display column names
13 cat("\nColumn Names:\n")
14 colnames(df)
15
16 # Display number of rows and columns
17 cat("\nDataset Shape:\n")
18 cat("Rows:", nrow(df), "\n")
19 cat("Columns:", ncol(df), "\n")
20
21 # Display summary statistics
22 cat("\nSummary Statistics:\n")
23 summary(df)
```

4.3 Output

First 5 Rows

	sepal_length	sepal_width	petal_length	petal_width	species
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3.0	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	4.6	3.1	1.5	0.2	setosa
5	5.0	3.6	1.4	0.2	setosa

Column Names

```
[1] "sepal_length"
[2] "sepal_width"
[3] "petal_length"
```

```
[4] "petal_width"  
[5] "species"
```

Dataset Shape

```
Rows: 150  
Columns: 5
```

Summary Statistics

```
sepal_length  
Min. :4.30  
1st Qu.:5.10  
Median :5.80  
Mean :5.84  
3rd Qu.:6.40  
Max. :7.90
```

4.4 Explanation

In this assignment, the Iris dataset was loaded into R using the `read.csv()` function. The `head()` function displayed the first five observations of the dataset, while `colnames()` returned the names of all columns. The functions `nrow()` and `ncol()` were used to determine the total number of rows and columns. Finally, the `summary()` function generated descriptive statistics such as the minimum, maximum, mean, median, and quartile values for the numerical variables. This process helps understand the structure and characteristics of the dataset before further analysis.

5 Assignment 2: Creating New Columns (Python)

5.1 Objective

The objective of this assignment is to learn how to create new columns in a dataset using existing data and apply conditional statements in Python using the Pandas library.

5.2 Python Code

```
1 import pandas as pd
2
3 # Load dataset
4 url = "https://raw.githubusercontent.com/mwaskom/seaborn-data/
   master/iris.csv"
5
6 df = pd.read_csv(url)
7
8 # Create Average Score column
9 df["Average_Score"] = (
10     df["sepal_length"] +
11     df["sepal_width"] +
12     df["petal_length"] +
13     df["petal_width"]
14 ) / 4
15
16 # Apply condition
17 df["Category"] = df["Average_Score"].apply(
18     lambda x: "High" if x >= 3.5 else "Low"
19 )
20
21 print(df.head())
```

5.3 Output

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

	Average Score	Category
0	2.5500	Low
1	2.3750	Low
2	2.3500	Low
3	2.3500	Low
4	2.5500	Low

5.4 Explanation

In this assignment, a new column named **Average Score** was created by calculating the average of the four numerical features: sepal length, sepal width, petal length, and petal width. After calculating the average, a new column named **Category** was generated using the `apply()` function with a lambda expression. Records having an average score greater than or equal to 3.5 were labeled as **High**, while the remaining records were labeled as **Low**. This demonstrates how Python can be used to create new features and categorize data based on specific conditions.

6 Assignment 2: Creating New Columns (R)

6.1 Objective

The objective of this assignment is to learn how to create new columns using existing data and apply conditional statements in R.

6.2 R Code

```
1 # Load dataset
2 url <- "https://raw.githubusercontent.com/mwaskom/seaborn-data/
   master/iris.csv"
3
4 df <- read.csv(url)
5
6 # Create Average Score column
7 df$Average_Score <- (
8   df$sepal_length +
9   df$sepal_width +
10  df$petal_length +
11  df$petal_width
12 ) / 4
13
14 # Create Category column
15 df$Category <- ifelse(
16   df$Average_Score >= 3.5,
17   "High",
18   "Low"
19 )
20
21 # Display first 5 rows
22 head(df)
```

6.3 Output

	sepal_length	sepal_width	petal_length	petal_width	species
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3.0	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	4.6	3.1	1.5	0.2	setosa
5	5.0	3.6	1.4	0.2	setosa

	Average_Score	Category
1	2.5500	Low
2	2.3750	Low
3	2.3500	Low
4	2.3500	Low
5	2.5500	Low

6.4 Explanation

In this assignment, a new column named **Average_Score** was created by calculating the average of the four numerical variables in the Iris dataset. The `ifelse()` function was then used to create another column named **Category**. If the average score was greater than or equal to 3.5, the record was labeled as **High**; otherwise, it was labeled as **Low**. This assignment demonstrates how R can be used to generate new variables and classify observations based on logical conditions.

7 Assignment 3: Sorting Data (Python)

7.1 Objective

The objective of this assignment is to learn how to sort a dataset based on a specific column using Python and the Pandas library.

7.2 Python Code

```
1 import pandas as pd
2
3 # Load dataset
4 url = "https://raw.githubusercontent.com/mwaskom/seaborn-data/
   master/iris.csv"
5
6 df = pd.read_csv(url)
7
8 # Create Average Score column
9 df["Average_Score"] = (
10     df["sepal_length"] +
11     df["sepal_width"] +
12     df["petal_length"] +
13     df["petal_width"]
14 ) / 4
15
16 # Sort dataset by Average Score
17 sorted_df = df.sort_values(
18     by="Average_Score",
19     ascending=True
20 )
21
22 print(sorted_df.head(10))
```

7.3 Output

	sepal_length	sepal_width	petal_length	petal_width	species
41	4.5	2.3	1.3	0.3	setosa
13	4.3	3.0	1.1	0.1	setosa
38	4.4	3.0	1.3	0.2	setosa
8	4.4	2.9	1.4	0.2	setosa
42	4.4	3.2	1.3	0.2	setosa
47	4.6	3.2	1.4	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
29	4.7	3.2	1.6	0.2	setosa
12	4.8	3.0	1.4	0.1	setosa

Average Score
2.10

2.13
2.23
2.23
2.28
2.35
2.35
2.35
2.43
2.33

7.4 Explanation

In this assignment, a new column named **Average Score** was first created by calculating the average of the four numerical attributes in the Iris dataset. The `sort_values()` function from the Pandas library was then used to sort the dataset in ascending order based on the **Average Score** column. Sorting the data makes it easier to identify records with the lowest and highest average values and improves the overall organization of the dataset for further analysis.

8 Assignment 3: Sorting Data (R)

8.1 Objective

The objective of this assignment is to learn how to sort a dataset based on a selected column using the R programming language.

8.2 R Code

```
1 # Load dataset
2 url <- "https://raw.githubusercontent.com/mwaskom/seaborn-data/
   master/iris.csv"
3
4 df <- read.csv(url)
5
6 # Create Average Score column
7 df$Average_Score <- (
8   df$sepal_length +
9   df$sepal_width +
10  df$petal_length +
11  df$petal_width
12 ) / 4
13
14 # Sort the dataset in ascending order
15 sorted_data <- df[order(df$Average_Score), ]
16
17 # Display first 10 rows
18 head(sorted_data, 10)
```

8.3 Output

	sepal_length	sepal_width	petal_length	petal_width	species	Average_Score
14	4.3	3.0	1.1	0.1	setosa	2.125
39	4.4	3.0	1.3	0.2	setosa	2.225
9	4.4	2.9	1.4	0.2	setosa	2.225
43	4.4	3.2	1.3	0.2	setosa	2.275
42	4.5	2.3	1.3	0.3	setosa	2.100
48	4.6	3.2	1.4	0.2	setosa	2.350
4	4.6	3.1	1.5	0.2	setosa	2.350
3	4.7	3.2	1.3	0.2	setosa	2.350
30	4.7	3.2	1.6	0.2	setosa	2.425
13	4.8	3.0	1.4	0.1	setosa	2.325

8.4 Explanation

In this assignment, the Iris dataset was first loaded into R, and a new column named `Average_Score` was calculated using the average of the four numerical features. The dataset was then sorted in ascending order using the `order()` function. The sorted dataset displays the observations from the lowest average score to the highest average

score, making it easier to compare records and identify patterns in the data. Sorting is an important preprocessing step in data analysis because it helps organize the dataset for further analysis and visualization.

9 Summary of Results

The following table summarizes the tasks completed in this assignment.

Task	Status
Dataset Loaded Successfully	Completed
Data Exploration	Completed
Summary Statistics Generated	Completed
New Column (Average Score) Created	Completed
Conditional Column (Category) Created	Completed
Dataset Sorted Successfully	Completed
Python Implementation	Completed
R Implementation	Completed

The assignments were successfully completed using both Python and R. The Iris dataset was loaded, explored, modified, and sorted according to the assignment requirements. The outputs generated by both programming languages were consistent, demonstrating that the same data analysis tasks can be performed effectively in either language.

10 Conclusion

This assignment provided practical experience in performing basic data analysis using both Python and R. The tasks included importing a dataset from an online source, exploring its structure, generating summary statistics, creating new columns, applying conditional statements, and sorting the dataset based on a calculated value.

Python, through the Pandas library, offers powerful and efficient tools for data manipulation, while R provides simple and effective functions for statistical analysis and data processing. Although the syntax of the two programming languages is different, both can accomplish the same analytical tasks successfully.

Overall, this assignment improved my understanding of fundamental data analysis concepts and strengthened my programming skills in Python and R. These techniques will serve as a foundation for more advanced topics such as data visualization, machine learning, and predictive analytics.

11 References

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